



Keeping the House in the Family: Non-Market Transfers and the Supply of American Homes



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Abstract

A large share of American homes change hands without a market. Using 144 million deed records covering the universe of recorded residential transfers from 2007 to 2023, this paper provides the first national deeds-based accounting of intra-family housing transfers. After stripping out trust retitling and same-person co-owner changes, families moved nearly 20 million homes between relatives over this period—one family transfer for every 3.6 market sales—almost always at a recorded price of zero. These transfers are not a staging step toward sale: three in five family-transferred homes have not sold on the open market a decade later. A simple model shows that when heirs inherit a below-market property-tax assessment along with the house, as permitted by California's Propositions 13, 58, and 193, families keep homes that the market values more. California's Proposition 19, which abruptly ended assessment inheritance for non-occupant heirs in February 2021, provides a natural experiment to examine the effect of this tax wedge on family transfers: family transfers surged 31% above counterfactual in the four months before the effective date and then fell to a new, persistently lower level: a donor-weighted synthetic control reveals a 2022–2023 decline of 33%, which decreases to 26% after adjusting for the elderly population. Net of California's own market-sale slump, the ~~missing~~ flow is roughly 58,000 deeds per year. The homes that exited the family channel have not yet surfaced as estate sales, indicating deferred successions rather than immediate listings. Our results suggest that family transfers are not outside the housing market but are one of its allocation mechanisms and property-tax rules can shift scarce homes from price-based matching to lineage-based retention.



never use "home"
use "house" instead

JEL classification: R21, R31, H22, D64

Keywords: housing supply, intergenerational transfers, bequests, property taxes, Proposition 19

1 Introduction

The housing market that economists study is the one with prices. Sales are recorded, indexed, and aggregated. Turnover, liquidity, and supply are all defined over transactions in which a willing buyer pays a willing seller. But county deed records reveal a second, often understudied, parallel circulation of houses that satisfies none of these definitions: family transfers. Between 2007 and 2023, American families recorded nearly 20 million transfers of residential property between relatives—a mother quitclaiming a bungalow to her son, siblings dividing a deceased parent’s house, a grandfather deeding a rental to a grandchild. For every 3.6 homes sold on the open market, one moved through this family channel. Almost none of these transfers involved money: nine in ten record no price at all. None involved a listing, an appraisal, or a price signal. The largest asset most households ever own is routinely allocated not by the market but by the family—and the deed records that document this allocation have gone largely unexamined.

This paper makes three contributions. First, it measures the family-transfer channel in housing. We construct a taxonomy of all 144 million recorded residential deed events in the United States from 2007 to 2023, separating arm’s-length market sales from inter-person family transfers, trust self-transfers, same-person co-owner retitling, and other non-market conveyances, using the names on the deeds themselves to distinguish a true intergenerational transfer from a household retitling its own home into a revocable trust or adding a spouse to a title. The family channel is large, stable, and systematically patterned: family transfers are relatively more common for older houses, for parcels with senior tax exemptions, and at the bottom of the value distribution—the signature of a bequest pipeline, not of disguised sales. Its geography is equally telling. The family share is highest in California, where one family transfer is recorded for every 1.6 market sales—2.3 times the national ratio—and where, not coincidentally, the property-tax code made dynastic holding most valuable. "makes"?

Second, it follows the transferred houses and asks, what happens to a home after it passes within a family? Because each parcel carries a permanent identifier, the deed records can be linked across successive owners to track the time to first market sale. The answer is bimodal. A minority of family transfers are settlement waypoints: among cross-surname family conveyances—the class that includes estate distributions—one in ten reaches the open market within a year, as heirs take title and cash out. But the majority are destinations. Ten years after a same-surname family transfer, 62 percent of homes have still never sold on the open market, a retention rate exceeding that of homes bought on the market by their owner-occupants and investors combined. Houses that

enter the family channel mostly stay there.

Third, it shows that tax policy shapes this allocation. Does the tax benefit of keeping a home in the family affect whether inherited homes reach the market? California offers the cleanest test for this question. Proposition 13 (1978) ties each parcel’s assessed value to its acquisition price. Propositions 58 (1986) and 193 (1996) let parents pass that assessed value to children along with the house. For instance, an heir could thereby inherit a Pasadena house worth \$1.5 million and pay property taxes on a 1980s basis—a subsidy available only if the family *kept* the house. The state’s Legislative Analyst’s Office estimated that 60,000 to 80,000 inherited properties escaped reassessment each year and observed, presciently, that the exclusion “appears to be encouraging children to hold on to their parents’ homes to use as rentals or other purposes instead of putting them on the for sale market” ([California Legislative Analyst’s Office, 2017](#)). Proposition 19, passed in November 2020 and effective February 16, 2021, abruptly ended this regime: the exclusion now survives only when the child occupies the home as a principal residence, and only up to a value cap; for rentals and second homes it vanished entirely ([California State Board of Equalization, 2025](#)). **I think it's well sold by property tax angle**

The deeds record the response with unusual clarity. In the four months between passage and the deadline, California families rushed to lock in the old basis: family transfers ran 31 percent—29,000 deeds—above a counterfactual built from California’s own seasonal pattern and other states’ contemporaneous growth. After the deadline they collapsed to a persistently lower level. In the 2022–2023 steady state, a donor-weighted synthetic control—California matched to a weighted combination of the other fifty states on its pre-reform trajectory—places family transfers 33 percent below synthetic California. Because California is a single treated unit, we assess this against a placebo distribution in which every other state is assigned the treatment in turn: California’s post-to-pre prediction-error ratio is the second largest of fifty-one ($p = 0.039$). Adjusting for the elderly population leaves a 26 percent gap, so we read the steady-state decline as 26 to 33 percent. Net of California’s contemporaneous market-sale slump, the missing flow is roughly 58,000 transfers per year, commensurate with the Legislative Analyst’s **estimate** of 60,000–80,000 tax-favored inherited properties annually. The timing is harder to mimic still: a monthly event study shows family transfers drifting up as the measure advances to the ballot, spiking in February 2021, and then settling 0.3 to 0.5 log points below their 2018–2019 baseline through 2022–2023—a within-year swing that annual data average away, consistent with the reform removing a large tax-motivated component of the channel.

happens

What happened to the homes that no longer pass between relatives? A simple model of the heir's keep-or-sell decision makes the prediction precise, and subtle. When the tax wedge disappears, the heirs who switch behavior are those who valued the house *less* than the market did and kept it only for the inherited tax basis—the marginal landlords. These homes need never generate a family deed at all: the succession can simply end in an estate sale. The family transfers that survive are selected—heirs who genuinely want the house, disproportionately to live in it—so the model predicts that post-reform family transfers should show a *lower* subsequent market-sale hazard, not because supply was not released but because the release happens upstream, in transfers that no longer occur. The conditional margins move in the predicted directions but are statistically unremarkable; the volume margin is where the houses are. The natural next question—do the missing homes surface as estate sales?—gets a clean answer in the data: not yet. Market sales by estate and trust sellers in California track other states through 2023. The transfers the reform eliminated were largely inter-vivos deeds executed years before death; the successions they anticipated have not yet occurred, and when they do, the homes will face market-value reassessment in the hands of any non-occupant heir. The reform's supply effect is real but deferred—an unusually clean prediction for future data to test. **is there a way to see if heirs value the house more than the market, to see if sentimental attachment to the house might play a role in not selling it?**

Our findings connect three literatures that have largely run on separate tracks. The economics of bequests—from strategic motives (Bernheim et al., 1985) to inter-vivos timing (McGarry, 1999) to estate-tax planning (Kopczuk, 2007) and the macroeconomic return of inheritance (Piketty, 2011)—has documented how wealth passes between generations, but in data where a house is a dollar amount, not a parcel with an address and a subsequent history. Housing wealth is the modal bequest and a central channel of intergenerational wealth correlation (Charles and Hurst, 2003); this paper supplies the asset-level record of how it moves. The lock-in literature shows that property-tax institutions and financing terms anchor owners in place—Proposition 13 lengthened California tenures (Wasi and White, 2005), assessment portability releases older movers (Ferreira, 2010), low mortgage rates immobilize borrowers (Fonseca and Liu, 2024), and property-tax design shapes allocation generally (Coven et al., 2024)—but its unit of observation is the living owner. This paper shows that in certain cases, the lock survives death. Quite separately, the transaction-tax literature (Best and Kleven, 2018; Hilber and Lyytikäinen, 2017; Kopczuk and Munroe, 2015; Slemrod et al., 2017) establishes that property transactions respond sharply to tax notches through timing and volume; the Proposition 19 deadline generates the same anticipation dynamics in transfers that have no price at all.

as to my point above, Gemini says you can

Closest in spirit, the misallocation argument descends from [Glaeser and Luttmer \(2003\)](#): when a scarce asset is allocated by queue—or here, by blood—rather than by willingness to pay, the loss is not a transfer but forgone surplus, and the houses withheld from the market are disproportionately the older, cheaper stock through which housing filters down to lower-income buyers ([Rosenthal, 2014](#)). To our knowledge, no prior academic study provides a national, deeds-microdata accounting of intra-family housing transfers or a causal evaluation of Proposition 19’s inheritance provisions; the nearest antecedents are an industry tabulation of inheritance deeds ([Delventhal, 2026](#)), measurements of the heirs’-property subset ([Burnett and Winters-Michaud, 2025](#); [Dobbs and Johnson Gaither, 2023](#)), and the Legislative Analyst’s own assessment-roll counts ([California Legislative Analyst’s Office, 2017](#)).

Two scope conditions discipline the welfare reading. First, family transfers are not waste as they carry sentimental value, insure family members, and economize on transaction costs, and when the heir values the home most, the family allocation is also the efficient one. The problem is the *tax wedge*: a tax code that pays heirs to hold homes they value below the market price converts a private gift into a social misallocation, larger where housing is scarcest. Second, the Proposition 19 estimates measure the response of recorded transfers in a single—if largest—state, whose acquisition-value assessment made the wedge uniquely large; the national counterfactual in Section 7 is correspondingly framed as bounds.

The paper proceeds as follows. Section 2 describes the deeds and the taxonomy. Section 3 establishes the national facts. Section 4 follows homes after transfer. Section 5 presents the model. Section 6 analyzes Proposition 19. Section 7 discusses magnitudes, and Section 8 concludes.

2 Data and the Anatomy of a Non-Market Transfer

2.1 Deeds

Property records come from CoreLogic’s Owner Transfer file ([CoreLogic, 2024](#)), obtained through an academic license, which compiles the universe of recorded deeds from county recorder offices. A deed is filed whenever the legal ownership of a parcel changes—not only when a house is sold. The same administrative trail that records a \$400,000 sale records a mother quitclaiming her house to her son for nothing. Market-based datasets (MLS listings, repeat-sales indices, mortgage originations) discard these zero-price events

by construction. Here they are the object of study.

The extract covers all fifty states and the District of Columbia. Coverage is dense from 2007 through mid-2024; we use 2007–2023 for annual statistics and treat 2024 as a partial year. Restricting to residential parcels and deduplicating to one event per parcel per recording date yields 148.7 million deed events (144.0 million in 2007–2023), linked through a permanent parcel identifier that follows each house across successive owners.

2.2 Classifying transfers

CoreLogic flags each record with a primary category—arm’s length or not—and an interfamily indicator constructed from the recorder’s consideration, deed type, and name fields. We sharpen these flags into a taxonomy using the buyer and seller names on the deed:

does \$10,000 follow precedent you can cite? can you also say your story is robust to a different cutoff?

1. **Market sale:** arm’s-length category, recorded price of at least \$10,000, not flagged interfamily (71.1 million events, 2007–2023).
2. **Same-person retitle:** the named principal is the same person on both sides of the deed—an owner adding or removing a co-owner, typically at marriage, divorce, or refinancing (7.1 million). No transfer between people occurs, so these are tracked separately and never counted as family transfers.
3. **Family transfer, same surname:** flagged interfamily, buyer and seller sharing a surname but not the same person, no trust involved (13.6 million). This is the conservative measure of inter-person family transfers—it misses, for example, transfers to a married daughter who changed her name, so it bounds the channel from below.
4. **Family transfer, other:** flagged interfamily without a surname match—cross-surname relatives and in-laws—plus estate/executor deeds (6.1 million).
5. **Trust self-transfer:** a trust appears on either side of the deed (12.1 million). The modal event is estate planning—a household retitling its own home into a revocable trust. Control does not change hands; tracked separately, never counted as family transfers.
6. **Other non-arm’s-length:** the remainder—divorce divisions across surnames, title corrections, nominal-price conveyances, and foreclosure-related instruments (34.0 million).

are these median prices adjusted for inflation somehow?

Table 1: Residential deed events by class, 2007–2023

Event class	N (2007–2023)	Zero/no price (%)	Median price ^a	Quitclaim share (%)	Absentee recipient (%) ^b
Market sale (arm’s length)	71,076,255	0.0	\$224k	2.0	26.8
Other non-arm’s-length	33,980,735	64.8	\$57k	17.8	55.9
Family transfer: same surname	13,610,459	90.3	\$97k	39.6	17.8
Trust self-transfer	12,074,757	96.1	\$130k	30.0	25.6
Same-person retitle (co-owner change)	7,123,579	92.3	\$63k	40.3	16.9
Family transfer: other interfamilial	6,073,863	88.3	\$155k	26.4	34.8
Estate deed (non-interfamilial)	57,697	9.6	\$167k	2.2	89.8
Family transfer: estate/executor	1,905	82.9	\$596k	8.6	35.6

Notes: Authors’ classification of CoreLogic Owner Transfer records, residential parcels, deduplicated to one event per parcel-date. ^aMedian of strictly positive recorded prices. ^bShare of events whose buyer mailing ZIP differs from the property ZIP, among events where both are observed.

Table 1 validates the taxonomy on its observable fingerprints, and the fingerprints are unambiguous. Family transfers look nothing like sales: 90 percent of same-surname transfers record a price of zero, against 0 percent of market sales (which require a price by construction) and 65 percent of the residual non-arm’s-length class; 40 percent travel by quitclaim deed—the instrument of gift—against 2 percent of market sales. Where a price is recorded at all in a same-surname transfer, the median is \$97,000, well under half the \$224,000 median market price: family “sales” happen at family discounts. Corporate buyers, who take 10 percent of market purchases, are essentially absent from same-surname transfers. These are gifts and bequests, not disguised transactions. One note on probate: formal estate keywords (“estate of,” executor, administrator) rarely appear in the recorded name fields, so explicitly labeled estate deeds are a small class; most succession conveyances travel as ordinary interfamilial deeds and are captured in the same-surname and cross-surname classes.

3 The Parallel Housing Market: National Facts

Fact 1: For every 3.6 market sales, one home moves within a family. Table 2 and Figure 1 report annual volumes. Between 0.9 and 1.6 million homes pass between family members every year (same-surname plus other interfamilial), against 3.0 to 5.8 million market sales: a family:market ratio between 0.25 and 0.34 in every year of the sample, with a 2007–2023 aggregate of 0.28. The conservative same-surname series alone runs 0.7 to 1.0 million per year. The channel is not a recession artifact: the ratio was as

Table 2: The parallel housing market: annual volumes

Year	Market sales	Family (same surname)	Family (broad)	Trust self-transfers	Family:market ratio (broad)
2007	3,906,268	884,266	1,068,273	518,750	0.27
2010	3,097,287	723,862	959,657	592,088	0.31
2013	3,793,643	803,373	1,132,061	680,645	0.30
2016	4,565,213	804,689	1,178,714	739,483	0.26
2019	4,956,337	819,739	1,315,281	741,526	0.27
2021	5,820,867	1,047,039	1,622,749	966,149	0.28
2022	5,076,718	792,479	1,284,736	901,638	0.25
2023	4,113,610	653,050	1,061,180	951,273	0.26

Notes: Residential deed events. “Family (broad)” = same-surname + other-interfamily + estate/executor classes; excludes trust self-transfers. Ratio = family (broad) / market sales.

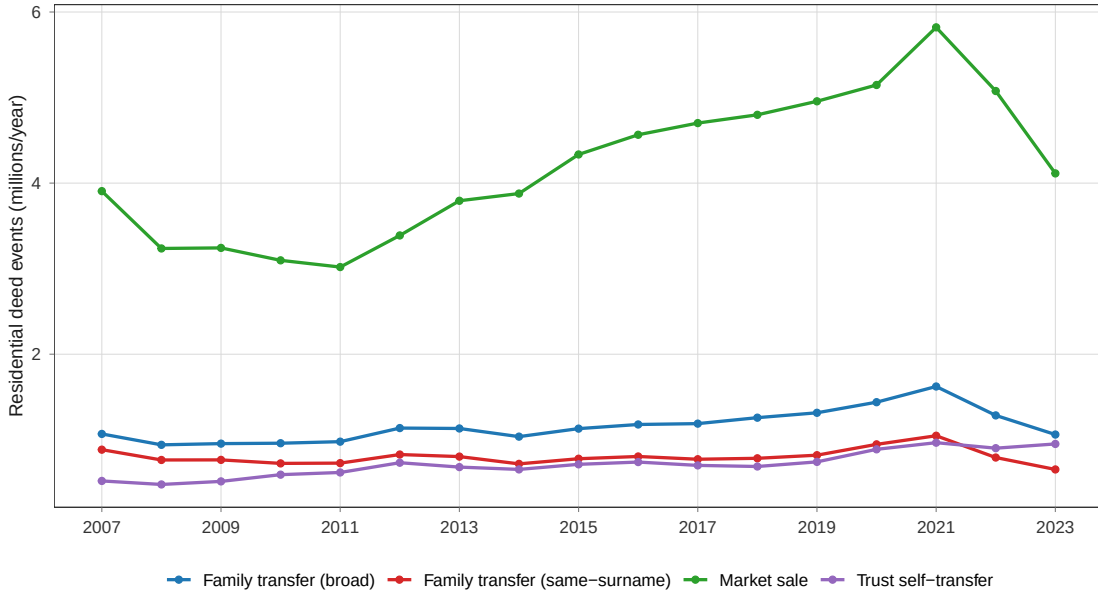


Figure 1: Residential deed events by class, 2007–2023. Family transfer (broad) combines same-surname, other-interfamily, and estate/executor classes; trust self-transfers are shown separately.

high in the 2009 bust (0.29) as in the 2021 boom (0.28). And the count is deliberately strict—it excludes the 12.1 million trust self-transfers, 7.1 million same-person retitles, and 34.0 million other non-market conveyances; adding them, the deed traffic outside the market rivals the market itself.¹

¹Restricting the denominator to existing homes (excluding new-construction first sales) raises the aggregate family:market ratio to 0.30. On a per-property rather than per-deed basis, the 19.7 million family-transfer deeds cover 16.5 million distinct parcels (1.19 deeds per parcel).

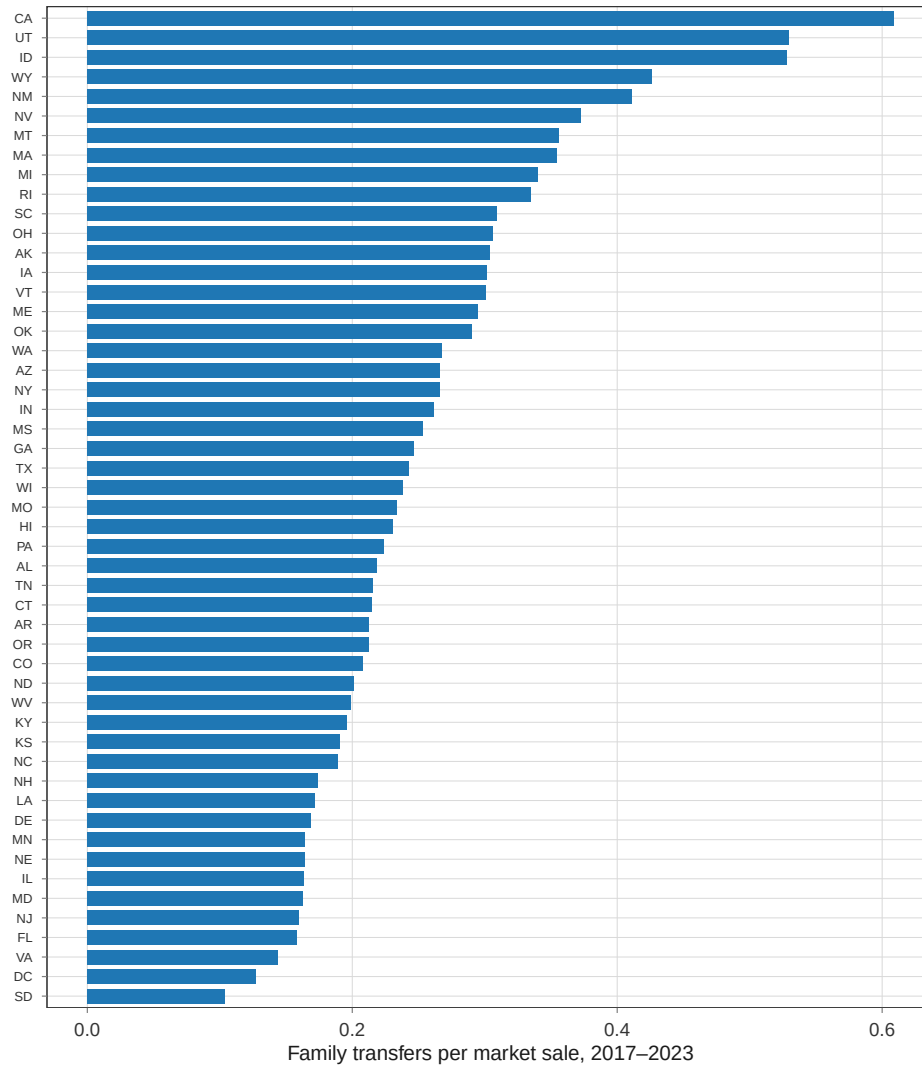


Figure 2: Family transfers (broad) per market sale by state, 2017–2023.

Fact 2: The family channel rises where holding is tax-subsidized. Figure 2 ranks states by the family:market ratio over 2017–2023. California leads the nation at 0.61—one family transfer for every 1.6 market sales, 2.3 times the pooled national ratio of 0.26—ahead of Utah and Idaho (0.53). California is exactly where acquisition-value assessment made the inherited tax basis most valuable, and it also accounts for 38 percent of all trust self-transfers in the country (2.2 million over seven years, more than its family transfers), the signature of an estate-planning industry built around property-tax basis. Cross-state correlations are not causal evidence—demography, land tenure, and recording practice all differ—but the cross-section puts the tax-subsidy hypothesis on the table, and Section 6 tests it within California.

Fact 3: Family transfers carry the old housing stock. Linking deed events to property characteristics (61 percent of 2017–2023 events match the current assessment snapshot; Appendix A), the family share of within-class traffic rises with structure age: family transfers are 20 percent of combined family-plus-market events for houses of ten to thirty years, 22 percent at thirty to fifty years, and 26 percent at fifty to seventy-five years. The family share is higher on parcels with senior tax exemptions (26 versus 22 percent)—the bequest pipeline made visible—and declines monotonically across the within-state value distribution, from 25 percent in the bottom quintile to 21 percent in the top. The houses that families keep are disproportionately the older, cheaper stock through which housing filters down to lower-income buyers (Rosenthal, 2014).

4 The Family Lock

What happens to a house after it enters the family channel? Because the parcel identifier links successive deeds, every transfer event can be followed to the property’s next arm’s-length market sale, if any. We take all events in the 2008–2018 cohorts—guaranteeing at least 66 months of follow-up before the June 2024 censoring date—and estimate Kaplan–Meier survival to first market sale, separately by event class: 8.5 million same-surname family transfers, 3.4 million other family transfers, 7.1 million trust self-transfers, 4.1 million same-person retitles, and 42.1 million market purchases as the benchmark.

Table 3 and Figure 3 show two distinct family destinies. The first is settlement: among cross-surname family conveyances—the class containing estate distributions among multiple heirs—10 percent reach the open market within twelve months and 15 percent within twenty-four, roughly double the market-purchase benchmark (5 and 9 percent). For a substantial minority, the family deed is the administrative final step before the house is sold.

The second destiny is the lock. Once a family-transferred home survives the settlement window, it becomes more immobile than the housing stock at large. Same-surname transfers start faster than market purchases (11.1 versus 8.9 percent sold within two years) but fall behind by year five (23.1 versus 23.7 percent) and keep falling behind: by year ten, 38 percent of same-surname family homes have returned to the market against 42 percent of market purchases. Conditional on passing the two-year settlement phase, 30 percent of family homes sell over the following eight years versus 36 percent of market purchases. The market-purchase benchmark, recall, includes investors and flippers; family homes are stickier than all of it. Three in five of the homes that passed between relatives

Table 3: Time to next open-market sale, 2008–2018 event cohorts

Event class	N	Sold on open market within (%)			
		12m	24m	36m	60m
Market sale (arm’s length)	42,055,369	5.2	8.9	13.7	23.7
Family transfer: same surname	8,467,604	6.9	11.1	15.4	23.1
Family transfer: other interfamily	3,426,450	10.1	14.7	18.9	26.5
Family transfer: estate/executor	665	9.9	14.3	41.2	45.9
Trust self-transfer	7,107,100	4.7	8.9	13.1	21.0

Notes: Share of events followed by an arm’s-length market sale of the same parcel within the stated horizon.

All cohorts observed at least 66 months before the June 2024 censoring date.

in 2008–2018 had not sold on the open market a decade later. The taxonomy passes an internal falsification along the way: same-person retitles—title changes with no transfer of control—track trust self-transfers almost exactly (9.1 versus 8.9 percent sold within two years), exactly as paperwork events should.

This is a descriptive fact, and selection is its content: families keep the homes they receive. Four caveats discipline its reading. First, the comparison pools states and cohorts; family transfers concentrate in older homes and in low-turnover states, so part of the gap is composition (a stratified version is planned for the next draft). Second, the two clocks measure different objects: a market purchase starts a fresh tenure, while a family deed often changes title without changing occupancy—a force that should, if anything, *raise* the family class’s sale hazard, since family deeds cluster near succession events. Third, some cross-surname family homes cannot sell for title reasons (heirs’ property), a lock of a different kind (Dobbs and Johnson Gaither, 2023). Fourth, trust self-transfers — pure retitling with no change of control—show survival close to the same-surname class, a reminder that long-tenure households are exactly the ones that do estate paperwork. Whether family retention reflects genuine attachment or a tax wedge is therefore exactly what the cross-sectional data cannot say—and what Proposition 19 can.

5 A Simple Framework

A two-line model clarifies why tax institutions matter for whether inherited homes reach the market, and organizes the California evidence.

A house yields a flow value u_m to the marginal market buyer. With property taxes levied at rate t on assessed value—and assessments reset to price at sale—the price capi-

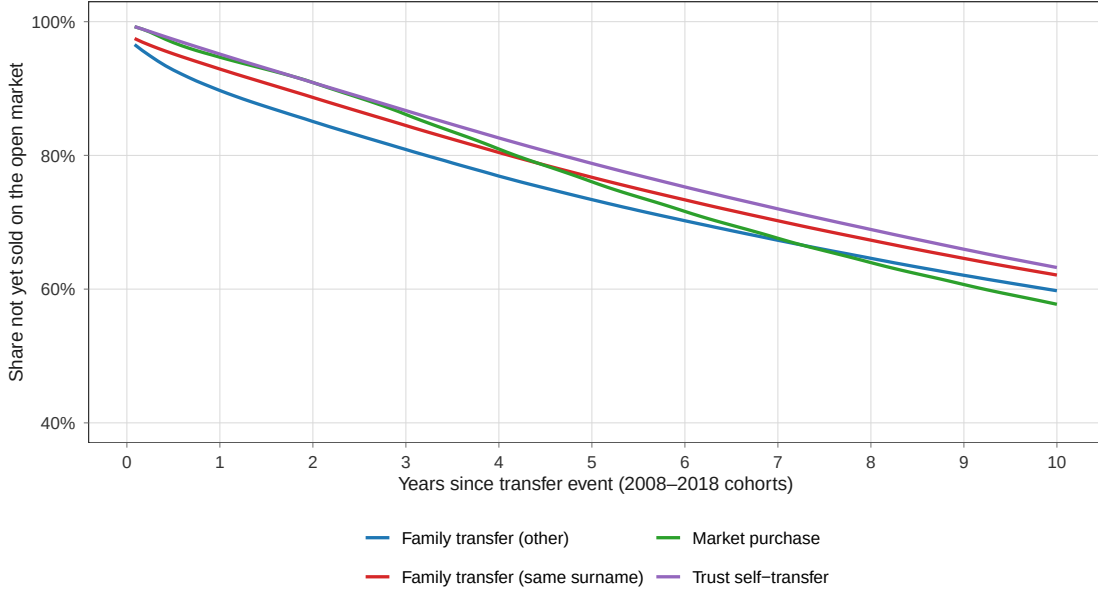


Figure 3: Kaplan–Meier survival to first open-market sale, 2008–2018 event cohorts, censored June 2024. The estate/executor class (n=915) is omitted.

talizes the tax burden in perpetuity:

$$p = \frac{u_m}{r + t}, \quad (1)$$

where r is the discount rate. Now consider an heir at the moment of succession who values living in the house (or renting it out) at flow u_h . If the family keeps the house, the heir pays taxes on assessed value A ; if the family sells, the heir receives p . Keeping the house is optimal if and only if

$$\frac{u_h - tA}{r} \geq p \iff u_h \geq u_m - t(p - A). \quad (2)$$

Equation (2) contains the paper’s economics. If succession triggers reassessment ($A = p$), the condition collapses to $u_h \geq u_m$: the house stays in the family only when the heir genuinely values it more than the market does, and the family channel is allocatively harmless. But if the heir *inherits the parent’s assessment* ($A = A_0 < p$), the family keeps the house whenever $u_h \geq u_m - t(p - A_0)$. Every heir with $u_h \in [u_m - t(p - A_0), u_m)$ retains a house that the market values more than she does. The wedge $t(p - A_0)$ —the annual tax saving from dynastic holding—is a price the tax code pays families to keep houses off the market. In California, where acquisition-value assessment dates to 1978

nicely said

and house prices have multiplied since, the wedge on a long-held home is measured in thousands of dollars per year, forever.

The key observational subtlety is that the *recorded family transfer is itself the outcome of the keep-or-sell choice*. When the family keeps, the county records a family deed; when it cashes out, the estate sells directly to a market buyer and no family transfer ever appears. The deed is endogenous to the tax regime. This shapes what an unanticipated removal of assessment inheritance for non-occupant heirs—which is what Proposition 19 did in February 2021—should do to each observable margin.

First, *retiming*: because a transfer completed before the deadline locks in A_0 , families with latent transfer plans rush to record deeds before the effective date, and recorded transfers collapse afterward (cf. the transaction-tax notch literature: [Best and Kleven, 2018](#); [Kopczuk and Munroe, 2015](#); [Slemrod et al., 2017](#)). Second, the *extensive margin*: with the wedge removed, every heir in $[u_m - t(p - A_0), u_m)$ switches out of the recorded family-transfer channel. If succession is imminent, the home can sell directly from the estate; if the pre-reform deed was advance tax planning, title simply remains with the parent until a later succession. The steady-state volume of family transfers falls by the mass of heirs in the wedge, and the supply release is therefore real but not necessarily contemporaneous. Third, *selection*: the family transfers that survive are those with $u_h \geq u_m$ —heirs who genuinely want the house, disproportionately owner-occupants, whom the surviving (occupancy-conditional) exclusion still favors. The model therefore predicts that post-reform family transfers should show a *lower* subsequent market-sale hazard and a lower absentee-recipient share—not because the reform failed, but because the marginal would-have-been-landlord heirs no longer take title at all.

6 Proposition 19: Death, Taxes, and the Return of Homes to the Market

6.1 Institutional background

Proposition 13 (1978) fixed each California parcel’s assessed value at its acquisition price, with growth capped at two percent per year; reassessment to market value occurs only at a change in ownership. Proposition 58 (1986) exempted parent-to-child transfers from this reassessment—for a principal residence of unlimited value plus up to \$1 million of assessed value of other property—and Proposition 193 (1996) extended the exemption to grandparent-grandchild transfers. A child could therefore inherit not just a house but its

decades-old tax bill, and keep that tax bill while renting the house out. The Legislative Analyst's Office estimated that 60,000 to 80,000 inherited properties escaped reassessment each year under this regime ([California Legislative Analyst's Office, 2017](#)).

Proposition 19, passed by voters on November 3, 2020, narrowed the exclusion sharply, effective February 16, 2021: it now applies only when the home was the parent's principal residence *and* the child makes it their own principal residence within a year, and only up to the inherited taxable value plus \$1 million (inflation-adjusted); the exclusion for rentals, vacation homes, and other non-primary-residence property was eliminated, with a carve-out for family farms ([California Legislative Analyst's Office, 2020](#); [California State Board of Equalization, 2025](#)). For a non-occupant heir, the reform sets $A = p$ in equation (2). County assessors reported a flood of parent-child filings ahead of the deadline.² The reform's other arm—expanded portability of assessments for movers aged 55 and over, effective April 1, 2021—affects market sales rather than family transfers; the designs below either difference it out with a market-sale placebo or use outcomes it does not touch.

Throughout this section, “family transfers” means the broad class (same-surname, cross-surname, and estate/executor deeds), measured monthly by state from the event panel; results including trust self-transfers are reported as robustness.

6.2 Retiming: the rush to beat the deadline

Figure 4 plots monthly California family transfers against a transparent counterfactual: California's 2019 volume in the same calendar month, scaled by contemporaneous family-transfer growth in the other fifty states and DC. The counterfactual thus inherits both California's seasonality and national shocks—COVID disruption, the 2020–2021 housing boom, the estate-planning surge—while excluding California from the donor pool. Before passage, it fits tightly: over 2018m1–2020m10, the mean absolute monthly error is 3.9 percent and the RMSPE is 5.7 percent. **this is the first occurrence of RMSPE in paper so spell it out here**

California tracks the counterfactual closely until November 2020. Between passage and the deadline, the two series tear apart: from November 2020 through February 2021, actual family transfers total 123,486 against a counterfactual of 94,617—an excess of 28,869 deeds, 31 percent above counterfactual, crammed into four months. February

²Sonoma County, for example, received roughly five times its typical volume of parent-child exclusion filings in the weeks before the February 15, 2021 deadline (North Bay Business Journal, February 2021); the Los Angeles County assessor accumulated a multi-year backlog of roughly 20,000 claims (Los Angeles County Auditor-Controller, 2025).

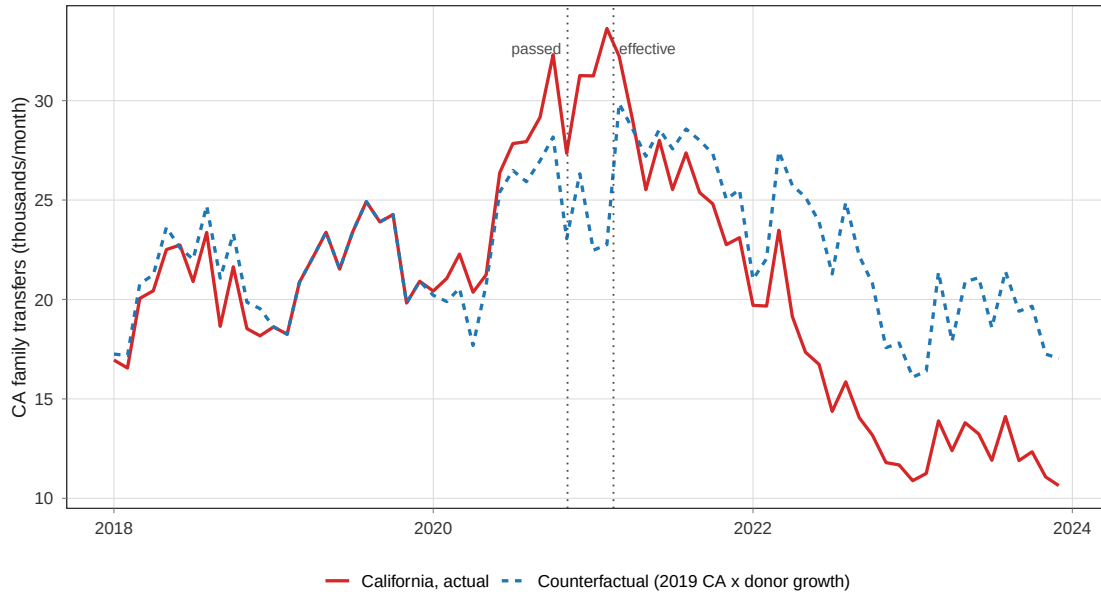
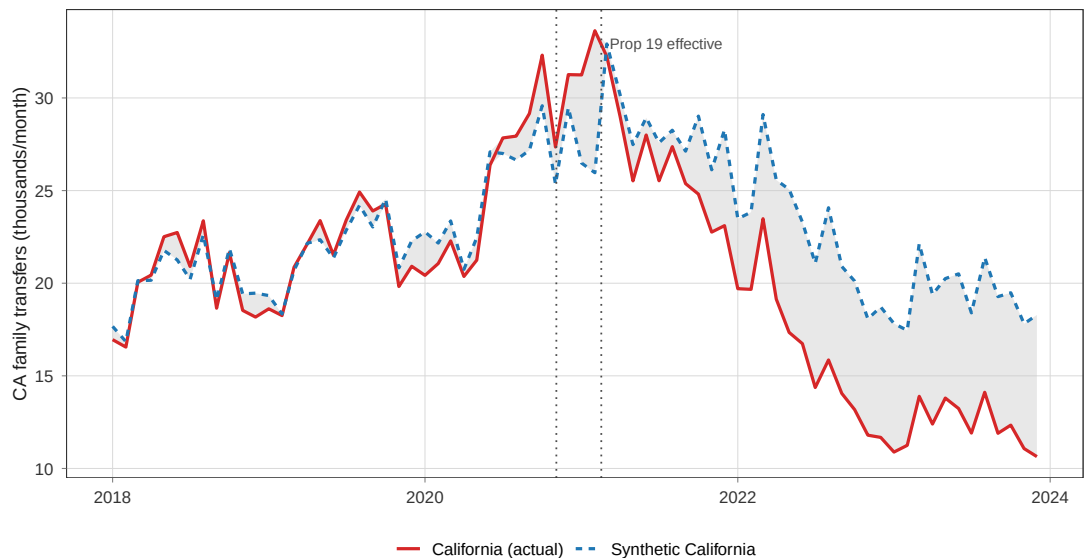


Figure 4: California monthly family transfers (broad class) versus counterfactual (CA 2019 same-calendar-month volume $\times$ donor-state growth). Dotted lines: November 3, 2020 (passage) and February 16, 2021 (parent-child provisions effective).

2021, ordinarily the year's quietest month, became the busiest family-transfer month in the 2018–2023 California series. This is the anticipatory bunching that notch designs recover for taxed market transactions (Best and Kleven, 2018; Slemrod et al., 2017), reproduced in transfers with no price: families re-timed successions to bequeath a tax basis. In leave-one-out placebo counterfactuals, California's anticipation gap is third of fifty-one in raw size ($p = 0.059$) but first after dividing each state's gap by its own pre-period RMSPE ($p = 0.020$). Because the response is timed to the day-specific deadline within a single state, no slow-moving confounder—pandemic, migration, the housing cycle—can mimic it.

6.3 The steady-state decline

Retiming alone would imply a transitory post-deadline trough followed by recovery. Instead, the gap persists and widens. From March 2021 through December 2022, actual transfers run 85,351 below counterfactual (16 percent); by 2023, California family transfers (147,441) are 41 percent below their 2018–2019 average (251,256)—while national volumes returned to trend. A donor-weighted synthetic control sharpens the comparison. Following Abadie et al. (2010), we build a synthetic California from a convex combination



Synthetic California = donor-weighted control (ADH). Pre-period fit RMSE 0.050. 2022–2023 gap ~33% below counterfactual; MSPE-ratio rank 2/51 (Fi

Figure 5: California monthly family transfers against a donor-weighted synthetic California (normalized-index match; Abadie et al., 2010). The series track closely before the reform and diverge after the deadline. Dotted lines: passage (November 3, 2020) and the parent-child effective date (February 16, 2021).

ask Nate or someone who has done research with synthetic controls, but my gut tells me you need more detail about the empirical approach, maybe a couple equations of the other fifty states and the District of Columbia, with weights chosen to match California's pre-reform family-transfer trajectory. Because California is the largest state—an outlier in the *level* of transfers, which forces a level-matched synthetic onto a single large donor—we match on each state's volume normalized to its own 2018–2019 mean. The synthetic reproduces California's pre-reform path to a root mean squared error of 5.0 percent (Figure 5); after the February 2021 deadline, California's family transfers run 33 percent below synthetic California through 2022–2023, and the gap lies at the extreme lower edge of the donor placebo distribution (Figure 6).

The difference-in-differences estimates in Table 4 formalize this. Using state-by-year counts for 2017–2019 versus 2022–2023 (the transition years 2020–2021 are excluded as contaminated by anticipation and retiming), with state and year fixed effects, log family transfers fall by 0.425 (s.e. 0.035) in California relative to other states—35 percent. The placebo column shows log market sales falling 0.163 (15 percent) over the same window, California's general post-rate-shock slump; netting it out leaves a 23 percent decline specific to family transfers, roughly 58,000 transfers per year against the 2018–2019 base. Reclassification into trusts cannot explain it: including trust self-transfers in the family aggregate yields a decline of 0.241 log points (21 percent).

at a minimum mention that you're using standard DID; with all the fancy new types of DID models, you might also have to justify why the Card and Krueger (1991) setup is adequate for this situation

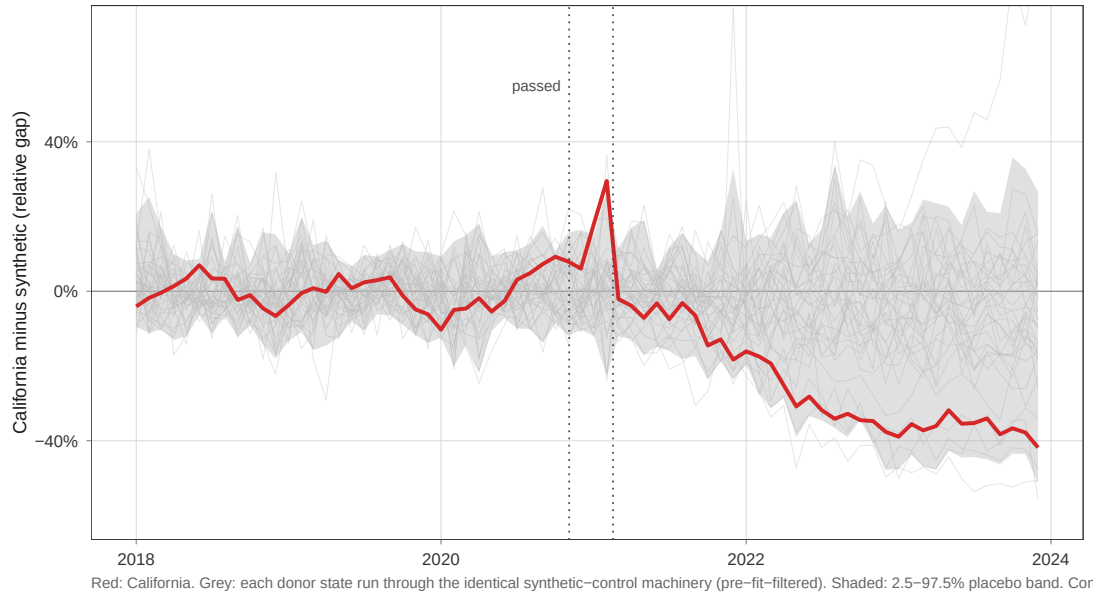


Figure 6: California’s relative gap from synthetic control (red) against the same gap computed for each donor state run as a placebo (grey lines), with a shaded 2.5–97.5 percentile band over donors that fit California comparably well before the reform. California lies at the extreme lower edge across 2022–2023.

Two pieces of evidence address the design’s limitations. First, because California is a single treated unit, we use the synthetic-control in-space placebo: each donor state is assigned the treatment in turn, and the ratio of its post- to pre-reform mean squared prediction error measures how extreme its post-period deviation is relative to its own pre-reform fit. California’s ratio is the second largest of fifty-one states—an exact randomization $p = 0.039$ (Appendix Figure A1)—sharper than, and free of the scaling choices in, the transparent counterfactual’s rank-based inference (Conley and Taber, 2011). A demographic adjustment brackets the magnitude rather than overturning it: re-estimating the synthetic control on family transfers per resident aged 65 and over (American Community Survey one-year estimates) leaves a 26 percent gap ($p = 0.137$). Because the eliminated deeds are predominantly inter-vivos tax-planning transfers rather than death-driven successions (Section 6.5), a denominator in the elderly population over-corrects; we read the steady-state decline as 26–33 percent, with 33 percent the central estimate. The anticipatory bunching above, timed to a single legislative deadline, is immune to differential mortality and aging by construction and remains the cleanest causal signature, with the steady-state decline as corroboration.

Second, timing. A monthly event study resolves what the annual specification hides. We regress log family transfers on California-by-month indicators with state and year-

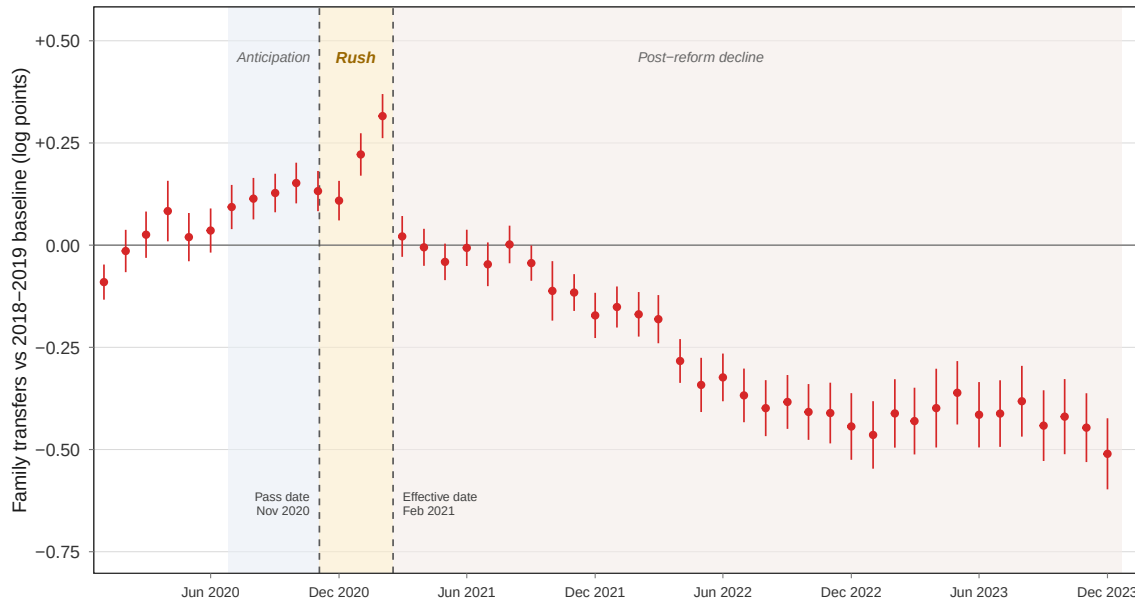


Figure 7: Monthly event study of California family transfers. Points are California-by-month coefficients from a regression of log family transfers on state and year-month fixed effects, with the 2018–2019 months pooled as the omitted baseline and the other fifty states and DC as controls. Whiskers are 95 percent state-clustered confidence intervals. Shaded regions mark the anticipation build-up (ballot to passage), the rush between passage and the February 16, 2021 effective date, and the post-reform period; dashed lines mark the passage and effective dates.

month fixed effects, the other fifty states and DC as controls, and the 2018–2019 months pooled as the omitted baseline. Figure 7 traces the whole path: California sits near its 2018–2019 baseline in early 2020, drifts upward through the anticipation period as Proposition 19 reaches the ballot and passes, spikes to +0.32 log points in February 2021—the deadline month—and then settles into a persistent decline of 0.3 to 0.5 log points through 2022–2023. An annual event study reports essentially zero for 2021 only because the pre-deadline rush and the post-deadline collapse cancel within the calendar year; at monthly resolution the two are plainly distinct, and the post-reform level shift is unmistakable. The missing annual flow, roughly 58,000 deeds, sits inside the Legislative Analyst’s 60,000–80,000 count of tax-excluded inherited properties per year, consistent with the reform removing a large tax-motivated component of the channel.

it's not a count, it's an estimated number, right?

Table 4: Proposition 19: volume, selection, and composition

Dependent Variables: Model:	log(n_fam) (1)	log(n_market) (2)	Sold within 24m (3) (4)		Absentee share (5)
<i>Variables</i>					
CA × Post	-0.4253 (0.0352)	-0.1626 (0.0293)	-0.0178 (0.0032)	-0.0104 (0.0019)	-0.0016 (0.0032)
Synthetic-control rank	2/51 (p=0.039)				
Ann. placebo rank	5/51 (p=0.098)				
Fam. − market log effect	-0.263				
Monthly pre-fit rank	1/51 (p=0.020)				
<i>Fixed-effects</i>					
State	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes			
Cohort month			Yes	Yes	Yes
<i>Fit statistics</i>					
Observations	255	255	1,836	1,836	1,836
R ²	0.98575	0.98961	0.84936	0.85147	0.90764

Clustered (State) standard-errors in parentheses

Cols (1)-(2): state-year counts, 2017-2019 vs 2022-2023. Cols (3)-(5): cohort cells (2017m1-2018m12 vs 2021m7-2022m6), weighted by cell size. Clustered (state) SEs are shown only for scale. Single-treated-state inference for the steady-state volume decline is the synthetic-control in-space placebo (MSPE-ratio rank 2/51, $p = 0.039$; Appendix Figure A1); the annual and monthly-counterfactual placebo ranks are reported as transparent alternatives. The monthly pre-fit rank is the 2022-2023 California gap divided by its 2018m1-2020m10 counterfactual RMSPE, ranked against leave-one-out placebo states.

6.4 Selection: who still keeps the house

add a sentence somewhere that pre-2020 trajectories for CA and rest of US are parallel, so it's fair to do triple differences

The model's third prediction concerns the transfers that survive. We take family-transfer cohorts from two clean windows—January 2017 to December 2018 (pre) and July 2021 to June 2022 (post, after the retiming turbulence)—and measure whether each transferred home reaches the open market within 24 months, a window that closes before the June 2024 censor for every cohort. Corporate recipients are excluded. The design compares California to other states, cohort month by cohort month (state and cohort-month fixed effects, weighted by cell size, clustered by state), with the identical regression run on market *purchases* as a placebo for California-specific housing-cycle shocks; the triple-difference nets the two.

The raw numbers tell the story (Appendix Figure A3). Before the reform, California family-transferred homes reached the market *more* often than the national average (16.1 versus 13.1 percent within 24 months)—California's inherited-basis regime made even hold-motivated transfers cheap to unwind later, and its hot market made selling attractive. After the reform, California converges toward the national rate (12.3 versus 11.1 percent). In the regression, the California post-reform effect is -1.78 percentage points; the market-purchase placebo shows -1.04 points, and the triple-difference is -0.74 points—a decline, as the selection logic predicts, not a rise. But candor about inference is required here:

the triple-difference sits well within the placebo-state distribution (permutation $p = 0.67$), and the absentee-recipient share is flat (-0.2 percentage points). The conditional margins move in the model’s direction and rule out the main alternative—had the missing transfers been settlement deeds, the surviving pool would have shifted sharply toward holders and the hazard would have fallen far more—but they do not carry independent statistical weight. The volume margin is where the evidence lives.

6.5 Where are the missing homes? A direct test

If the missing deeds were near-succession transfers, they should reappear as market sales by estates and trusts. If they were advance tax-planning deeds, they need not surface immediately: the parent simply keeps title until death. The deed names permit a direct test of the near-succession channel: market sales whose seller carries an estate, executor, or trust designation. The answer, so far, is that they have not reappeared. Estate- and trust-seller market sales in California fall 8 percent relative to donor states between 2017–2019 and 2022–2023 (permutation $p = 0.71$); relative to California’s overall market slump they *gain* 8 log points—their share of California market sales rises from 28.4 to 31.9 percent—but similar share gains occur elsewhere (permutation $p = 0.63$).

This null is informative as it discriminates between two readings of the volume decline. If the eliminated deeds had been transfers of homes whose succession was imminent—parents recently deceased, estates in settlement—the market should have received those homes within the 2022–2023 window. It did not. The eliminated deeds were instead predominantly *inter-vivos* transfers, executed while the parent was alive precisely to lock in the tax basis years or decades ahead of succession. Absent the tax motive, the parents simply keep title, and nothing observable happens until they die. The reform’s effect on market supply is therefore real but deferred: every one of these postponed successions now ends with reassessment at market value in the hands of any non-occupant heir, and equation (2) says more of those homes will be sold when the time comes. The deeds data will eventually record whether they are—a sharp, falsifiable prediction this paper leaves armed.

6.6 Threats to identification

California is a single treated state, so conventional clustered standard errors understate uncertainty regardless of the number of control states. Table 4 reports them for scale, not as the basis for inference. We instead rely on the synthetic-control in-space placebo—each

as to my point above, Gemini says you can test whether Prop 19 volume was driven by less absentee-recipient family transfers vs. local recipient transfers. if local transfers were unaffected but absentee transfers fell, it shows the tx wedge was leaving behind the heirs with sentimental attachment

donor state is assigned the treatment in turn and California's post-to-pre prediction-error ratio is ranked against the resulting distribution (second of fifty-one, $p = 0.039$)—together with Conley–Taber intervals built from the same placebo gaps. Placebo-state ranks for the transparent counterfactual, scaled by each state's pre-period RMSPE so that poorly fit placebos do not mechanically dominate, are reported as a corroborating alternative in Appendix B.

Five substantive threats to identification deserve mention.

- *COVID and the 2020–2021 turbulence*: the bunching counterfactual absorbs national pandemic shocks by construction, and its pre-period fit is checked over 2018m1–2020m10; the steady-state DiD excludes 2020–2021 entirely; and the placebo run on market sales nets out California-specific cycle effects.
- *Differential mortality and migration*: family transfers are driven by deaths and aging, and California's cumulative COVID mortality was below the national average while elevated mortality elsewhere pulled successions forward—both forces could depress California's 2022–2023 family transfers relative to donors through channels the market placebo does not absorb. We address this directly: re-estimating the synthetic control on family transfers per resident aged 65 and over (American Community Survey one-year estimates) leaves a 26 percent gap ($p = 0.137$). Because the eliminated deeds are predominantly inter-vivos transfers rather than death-driven successions, a denominator in the elderly population over-corrects, so we treat 26 percent as a conservative floor and the unadjusted 33 percent as the central estimate; the deadline bunching is immune to this channel by construction.
- *The pulled-forward stock*: some of the post-period shortfall reflects transfers executed early during the rush rather than abandoned. The paper's own numbers bound this: even if every one of the 28,869 excess deeds was borrowed from the March 2021–December 2022 window, retiming would account for at most a third of that window's 85,351-deed shortfall—and none of the 2023 decline, which is the largest. yeah, I wondered about that. you do great now at anticipating objections
- *Placebo contamination*: two forces push California market sales up after 2021 — the portability arm of Proposition 19 itself and the predicted release of inherited homes—which makes the market-sale placebo an imperfect cycle control; together with the evident acyclicity of family-transfer volumes (their ratio to market sales was as high in the 2009 bust as the 2021 boom), this brackets the steady-state effect between the placebo-adjusted estimate and the unadjusted one.

as to my point above, Gemini says you can test whether Prop 19 volume was driven by less absentee-recipient family transfers vs. local recipient transfers. if local transfers were unaffected but absentee transfers fell

- *Recording substitution*: families might route around the reform via instruments that escape the interfamily flag—LLCs, irrevocable trusts. The trust-inclusive estimate bounds substitution into trusts; substitution into opaque vehicles would have to be implausibly large to overturn the net decline, though it remains a reason to read the recorded magnitude as a lower bound on the behavioral response.

7 Magnitudes

Two back-of-envelope calculations situate the estimates.

California. The reform removed roughly 58,000 recorded family transfers per year (23 percent of a 251,000 baseline, net of the market placebo; the gross estimate is 87,000). Each deed is not a distinct house —family-held parcels average 1.19 family deeds over the window—so the flow of affected *properties* is somewhat smaller, squarely inside the Legislative Analyst’s 60,000–80,000 annual count of tax-excluded inherited properties. What the reform did to those homes is, on the direct evidence of Section 6.5, mostly deferral: the inter-vivos transfers that stopped were years ahead of succession, so the homes remain with their original owners and will face market-value reassessment—and the keep-or-sell test of equation (2) without the wedge—when succession comes. The eventual stakes are visible in the stock: at the pre-reform pace, the eliminated flow compounds to half a million California homes per decade whose succession now happens at market-value taxes rather than inherited ones, against a state market currently transacting 340,000 sales per year. How much of that converts to listings is the sharpest open question the data will answer as successions mature.

The nation. The wedge $t(p - A_0)$ requires an assessment gap, and California’s acquisition-value system—half a century old, in the nation’s most appreciated market—is the extreme. Most states reassess regularly, shrinking $p - A_0$ toward zero and with it the tax subsidy to dynastic holding; the federal subsidy that remains (stepped-up basis at death under IRC §1014, versus carryover basis for inter-vivos gifts under §1015, with estate taxes neutered by an exemption the 2017 tax act doubled) rewards dying with the house but not keeping it afterward. The national family channel—1.1 million homes per year against 4.1 million market sales in 2023—therefore reflects mostly non-tax motives, and no national reform should be expected to move a quarter of it. The right national reading runs the other way: nearly 20 million homes entered family hands off-market over 2007–2023, three in five still



unsold a decade on, and the California experiment shows that where assessment law subsidizes dynastic holding, the tax-motivated share of the channel is large. For the states and localities now debating assessment caps and inheritance carve-outs, the estimate of what such provisions do to the circulation of homes is no longer hypothetical.



8 Conclusion

Economists measure housing markets through prices, but for every 3.6 American homes that sell on the open market, another changes hands without one. This paper’s contribution is to make that parallel circulation visible—nearly 20 million family transfers from 2007 to 2023, nine in ten at a recorded price of zero, three in five still in family hands a decade later—and to show that it responds in market-like ways to tax incentives: bunching at deadlines and contracting sharply when the subsidy to dynastic holding ends. The family is not outside the housing market; it is a housing-market institution, one that allocates by birth and that policy has been subsidizing for nearly half a century in the largest state.

The agenda this opens is concrete. Longer post-reform horizons will reveal how many of the missing California successions surface as estate sales, closing the bounds in Section 7. Linking transferred parcels to rental listings and occupancy records would quantify the under-utilization of family-held homes directly. And the distributional question looms: family transfers transmit housing wealth within families that have it—[Charles and Hurst \(2003\)](#) document how tightly wealth correlates across generations—while withholding the filtering stock from families that do not ([Rosenthal, 2014](#)). Whether the family channel narrows or widens the gap between those who inherit homes and those who must buy them is, on the evidence here, a question tax policy helps decide.

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Appendix

A Data Construction Details

Source and scope. The CoreLogic Owner Transfer file compiles recorded deeds from county recorder offices in all fifty states and the District of Columbia. Our extract contains all record types—deeds of sale, quitclaims, intra-family conveyances, trust transfers, and foreclosure instruments. Coverage density is stable from 2007 onward; recording lags truncate the file around August 2024. All annual statistics use 2007–2023; the Proposition 19 cohort analyses use outcome windows that close by June 30, 2024.

Deduplication. Multi-parcel transactions and re-recorded corrections generate multiple rows per economic event. We keep one record per parcel-date pair, preferring arm’s-length category records and, within category, the highest recorded consideration.

Classification. The taxonomy in Section 2 is implemented with the recorder-derived interfamily indicator, the arm’s-length category code, deed type, recorded consideration, and string features of the buyer and seller name fields (surname equality; exact same-person matches on either buyer slot, which define the same-person retitle class; trust keywords; estate, executor, probate, and administrator keywords). Name fields are populated for the large majority of records (the seller surname is non-missing for 84 percent of interfamily events and the buyer surname for 66 percent); missing names can only push true family transfers into the “other interfamily” class, never into the market class, so the same-surname series is a strict lower bound on inter-person family transfers.

Absentee recipients. A recipient is absentee when the buyer mailing ZIP on the deed differs from the property’s situs ZIP, among records where both are observed (77 percent of events in 2007, rising to 98 percent by 2023). Within-state comparisons are unaffected by the secular improvement in coverage; difference-in-differences designs absorb it in time effects.

B Robustness Checks

Trust-inclusive volume DiD. Re-estimating the Proposition 19 volume difference-in-differences with trust self-transfers included in the family aggregate yields a $CA \times Post$

coefficient of -0.241 (s.e. 0.027), a 21 percent decline, confirming that the steady-state drop in family transfers is not an artifact of reclassification between direct family deeds and trust vehicles.

Synthetic-control in-space placebo. The main-text inference for the steady-state decline assigns the treatment to each donor state in turn and ranks California’s post/pre mean squared prediction error ratio in the resulting distribution; California is second of fifty-one (Figure A1, exact randomization $p = 0.039$).

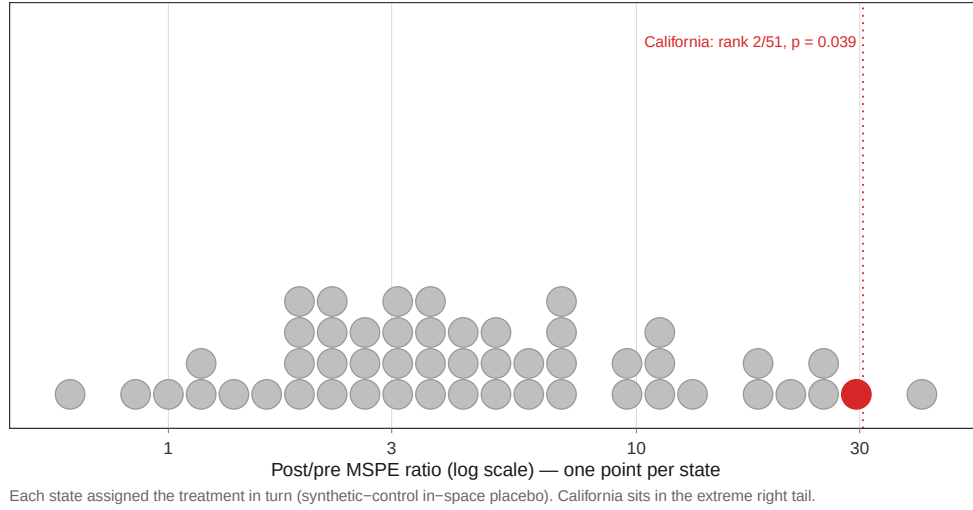


Figure A1: Synthetic-control in-space placebo: the post/pre mean squared prediction error ratio for each state assigned the treatment in turn. California (red) is second of fifty-one ($p = 0.039$).

Permutation and pre-fit-adjusted ranks. As transparent alternatives to the synthetic control, each donor state is assigned the treatment in turn, each design is re-estimated, and California’s coefficient is ranked in the placebo distribution. For the annual family-volume DiD, California ranks 5th of 51 ($p = 0.098$); for the placebo-netted estimate, 9th ($p = 0.176$). The monthly counterfactual also reports ranks after scaling each state’s policy-window gap by its own 2018m1–2020m10 RMSPE. Under the 2019 same-month baseline, California ranks 3rd of 51 in raw anticipation, February, and 2022–2023 gaps ($p = 0.059$), but 1st of 51 after pre-fit scaling in all three windows ($p = 0.020$). Using a 2018–2019 same-month baseline leaves the pre-fit-adjusted ranks unchanged. For the sold-within-24-months triple difference, $p = 0.67$; for the estate/trust-seller market-sale DiD, $p = 0.71$ (levels) and $p = 0.63$ (net of overall market).

Gemini says I'm right on with this comment; to concede that N=1 treated units is why you're relying onthe synthetic controls

Gemini says I'm right on with this comment; to concede that N=1 treated units is why you're relying onthe synthetic controls and Conley-Taber in-space placebos as the preferred estimate

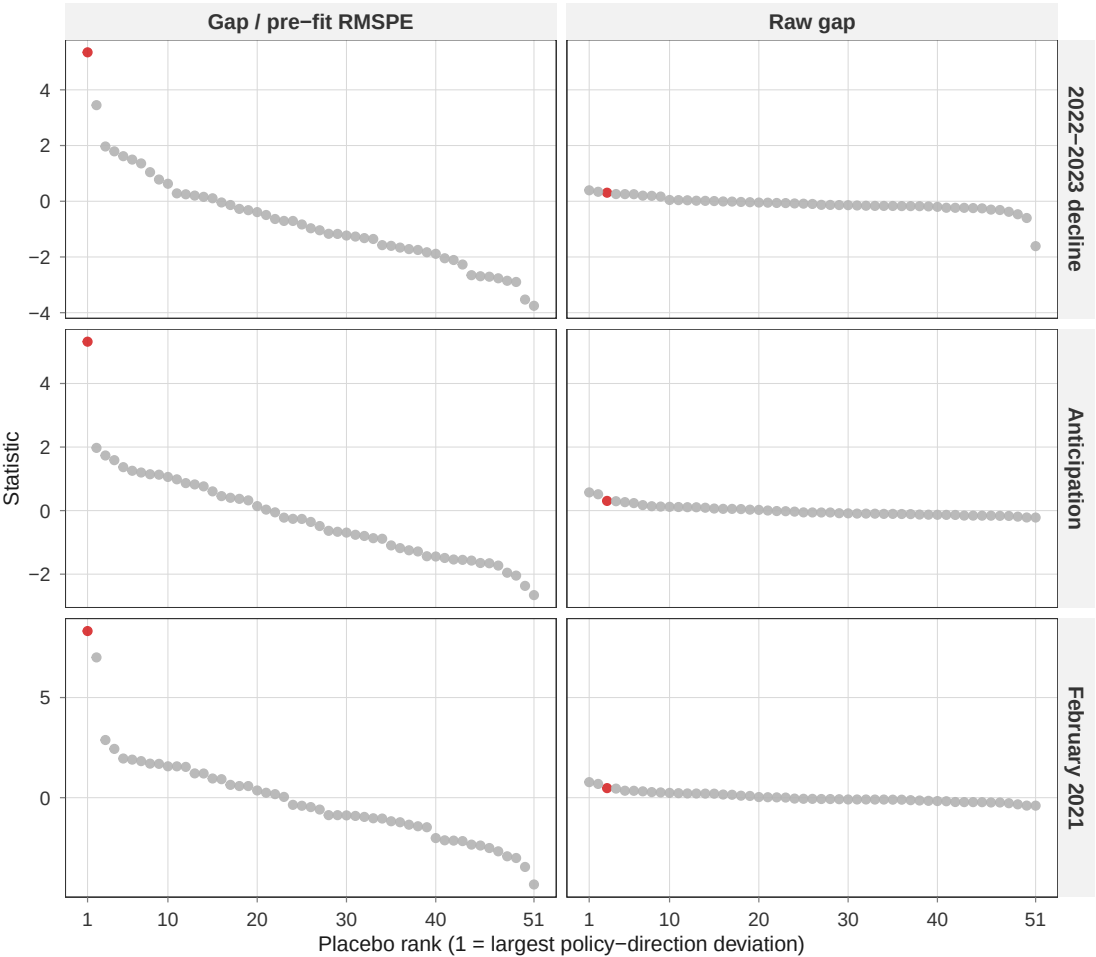


Figure A2: Placebo ranks for the monthly Proposition 19 counterfactual. Red points mark California; gray points assign the same policy window to each donor state in turn. Raw gaps rank California third in the baseline comparison, while gap divided by pre-period RMSPE ranks California first.

Selection: sold within 24 months by cohort. Figure A3 plots the conditional-margin series discussed in Section 6; it is relegated here because the triple difference is statistically unremarkable ($p = 0.67$) and the identified action is on the volume margin.

Raw difference-in-differences means. Table A1 reports the unadjusted means underlying the sold-within-24-months estimates.

okay, so here's where you graph parallel trends for DID

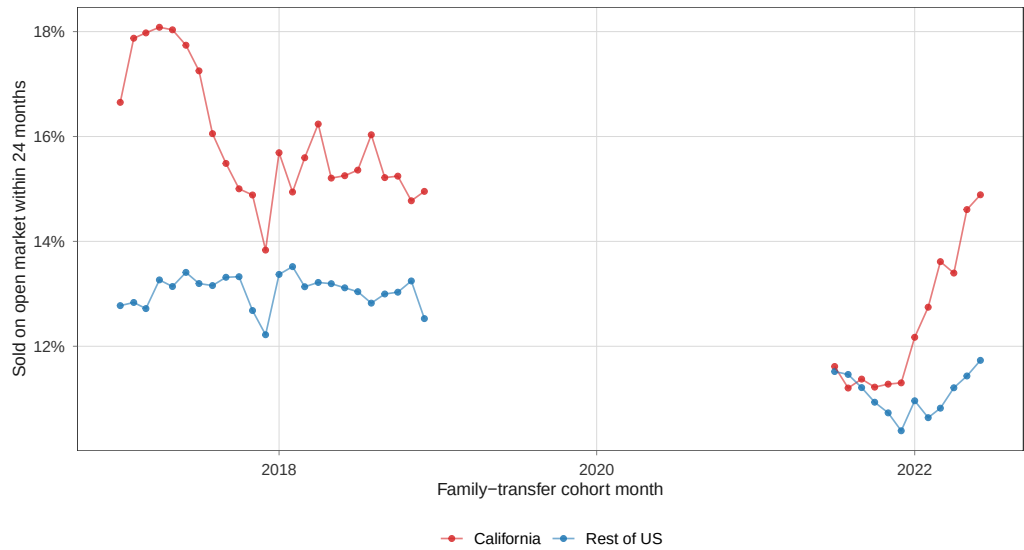


Figure A3: Share of family-transferred homes sold on the open market within 24 months, by cohort month. The plotted windows exclude the transition period around Proposition 19; lines are not connected across the omitted gap.

Table A1: Sold on open market within 24 months: raw cohort means

Cohort group	Region	Pre (2017m1–2018m12)	Post (2021m7–2022m6)	Change
Family transfer	California	16.1% (n=470,163)	12.3% (n=247,417)	–3.8pp
Family transfer	Rest of US	13.1% (n=1,793,673)	11.1% (n=1,118,028)	–2.0pp
Market purchase	California	6.2% (n=788,627)	4.7% (n=421,807)	–1.5pp
Market purchase	Rest of US	6.9% (n=7,722,553)	6.5% (n=4,495,875)	–0.4pp

Notes: Non-corporate recipients only. Family transfer = same-surname, cross-surname interfamily, and estate/executor classes (same-person retitles excluded). The difference-in-differences for family transfers is –1.8pp; for market purchases, –1.1pp; the triple difference is –0.74pp (regression counterpart in Table 4; permutation $p = 0.67$).